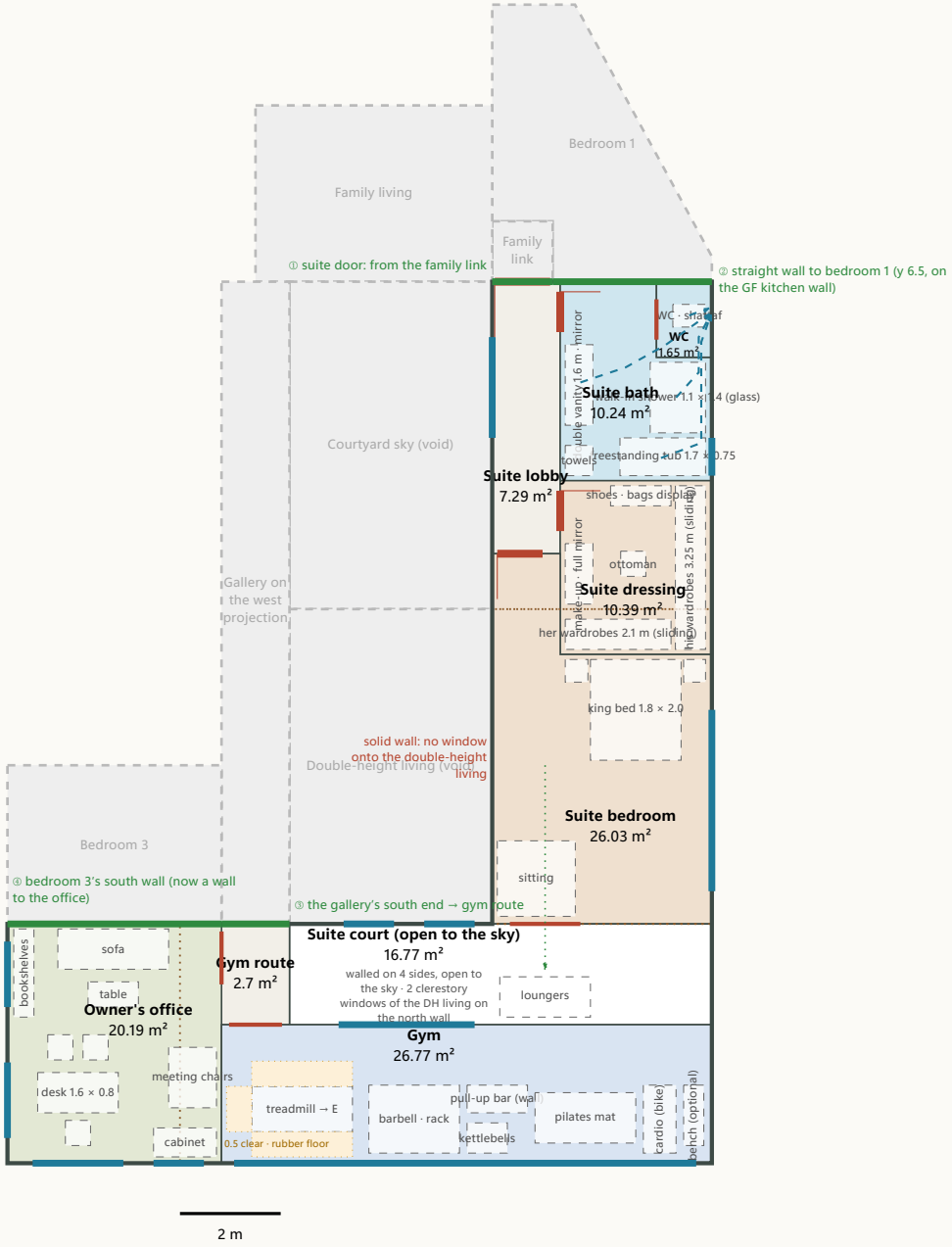
First floor · the L: owner's office → gym route → gym → court → master suite

Taif / Palm House · first floor V6.0 · CSV coordinates (x east, y south) · wall centre lines · north up

First floor V6.0 · part 2 of 2 (part 1: studies/ff_family.typ) · owner's office, gym (treadmill at the west end), suite court and master suite



Rooms

Room	Axes w × d (m)	Axes m²	≈ clear m²
Suite lobby	1.35 × 5.4	7.29	5.94
Suite bath	polygon	10.24	8.79
Suite WC (closet)	1.1 × 1.5	1.65	1.09
Suite dressing	3.01 × 3.45	10.39	9.01
Suite bedroom	polygon	26.03	23.56
Suite court (open to the sky)	8.39 × 2	16.77	–
Gym route	1.35 × 2	2.7	2.03
Owner's office	4.25 × 4.75	20.19	18.16
Gym	9.74 × 2.75	26.77	23.96
Zone		**122.04**	

Clear ≈ axes minus half a wall on every edge (250 mm outer, 200 mm inner, nothing on open edges), corners ignored.

✓ All rooms inside the zone · no overlaps · no unassigned area · furniture inside its room, not overlapping, out of the lobby route and the gym walkway · treadmill 0.5 m behind and each side, gym door free · door swings clear · every room has a door reached from the lobby (or its room) · doors from the family part on the interfaces · lobby ≥ 1.0 m clear · no window onto the double-height living

Key numbers

- **Suite:** lobby 7.29 · bath 10.24 + WC closet 1.65 · dressing 10.39 · bedroom 26.03 = **55.6 m² indoors** + the suite court 16.77 m².
- **Gym** 26.77 m² · **suite court** 16.77 m² · **gym route** 2.7 m² · **owner's office** 20.19 m².
- **Walked:** suite door → bedroom door ≈ 5.4 m · bedroom door → bath door ≈ 5.5 m · bed → court ≈ 3.9 m.
- **Interfaces with the family part (green, fixed):** ① suite door at the end of the family link; ② the straight wall to bedroom 1 on y 6.5; ③ the gallery's south end, where the gym route starts; ④ bedroom 3's south wall (a wall to the office).

Plan: red = doors (thin = sliding / no leaf) with swing line · blue = windows · blue dashed = drains · green = interfaces · brown dotted = ground floor below · grey = the family part (fixed) · dashed boxes = furniture and fixtures.

Master suite, suite court, gym and office · rooms, drains, walls, questions

Suite lobby · 1.35 × 5.4 m

- The suite door (from the family link) opens into a distributor: bath, dressing and bedroom each open off it; nobody walks through the dressing. A window onto the courtyard on its west wall.
- **Bent entry:** the bedroom door is at the lobby's south end, round the corner of the dressing: the bed is not seen from the suite door.

Suite bath (luxury, compact) · 11.89 m² incl. its WC closet

- **Freestanding tub** along the south wall by an east window; **walk-in glass shower** on the east wall; **double vanity** on the west (lobby) wall facing the window light; **towel cupboard**; open floor in the middle.
- **Door** from the lobby at the north end of the west wall, opening in, swing clear; the north wall stays clear as the path from the door to the WC closet.
- **WC in its own closet** (1.1 × 1.5 m, sliding door) in the north-east corner, at the drain stack, with a shattaf.

Suite dressing (walk-in) · 3.01 × 3.45 m

- Door from the lobby at the north end of its west wall. **His wardrobes** along the east wall and **her wardrobes** along the south wall, both sliding; a shoe and bag display on the north wall; a make-up table with a full-length mirror on the west wall; an ottoman. No window: good lighting inside the wardrobes.

Suite bedroom · 26.03 m²

- **King bed** 1.8 × 2.0 on the dressing wall, facing south to the suite court; bedside tables; a sitting corner on the solid west wall.
- **West wall solid:** the double-height living is behind it, no window (checked). Windows east and the sliding door to the court (south).

Suite court · 8.39 × 2 m, walled on four sides, open to the sky

- 16.77 m², from the bedroom only (sliding door). Walled on all four sides: north the double-height living (clerestory windows) and the suite bedroom, south the gym's north wall (high obscured windows), west the gym route, east the outer wall. Fully private; it works as a light well for the clerestory windows.
- **Clerestory windows** of the double-height living (in its south wall y 19.25): their sill stays above eye level from the court floor (or obscured glass). The gym's roof edge ends in a **railing, not a parapet**, to keep the winter sun on them (light check on design_ff.pdf).
- **Open to the sky:** its own floor drain and waterproofing on the men's majlis roof slab; the clerestory windows and the bedroom door open for air in summer.

Gym · 9.74 × 2.75 m, 26.77 m² · a rectangle

- **One straight bar** along the south facade, sliding door from the gym route at its west end; it does **not** touch the suite bedroom (the suite court lies between). Windows south and high windows north onto the suite court (obscured).
- **Treadmill at the west end**, away from the master suite: 0.9 × 2.0 m along the gym's length, the user facing east, **0.5 m clear behind** (to the office wall) **and on each side**; a **dense rubber floor** under it and its clear zones, **anti-vibration pads** under its feet. Along the gym's depth it would fill 2.5 of the 2.525 m clear depth and block the way from the door, so it runs along the length.
- **Along the south wall to the east:** barbell + rack, kettlebells, pull-up bar (wall), pilates mat, one cardio machine (bike) and an **optional bench at the east end**. A clear walkway runs along the north wall from the door to the east end (checked: nothing in it, the door landing free).

Owner's office · 4.25 × 4.75 m, 20.19 m² · gym route 1.35 × 2 m

- An indoor room over the GF foyer / guest passage and the men's majlis; its west part is on the outward cantilever, a canopy over the guest entrance.
- Sofa, coffee table and bookshelves in the north half; desk with two visitor chairs, a meeting corner and a cabinet in the south half. Windows west (two) and south (two).
- **Door** (sliding, lockable) from the gym route, just past the end of the gallery. The gym route serves the office and the gym.

Drains (blue dashed) · no east duct

- **Suite bath:** all fixtures drain east in a **sunken bath floor** to one **stack on the outside face of the east wall**, down to the east-yard manhole where the kitchen already drains. Nothing hangs in the GF kitchen ceiling; the hood on the GF east wall stays clear of the stack (south of y ≈ 7.5).
- No drain over the master bedroom. The suite court drains to its own floor drain.

Walls (engineer)

- **On GF lines:** y 6.5 (bedroom 1 / suite: the GF kitchen's north wall), the outer walls, x 8.95 (courtyard / double-height living). The suite bedroom spans the GF kitchen / living wall (y 13.0) with no wall above it.
- **Off the GF lines:** x 10.3, y 10.45, x 12.211 and y 8.0, y 11.9 and y 13.9, all over the GF kitchen; on the men's majlis roof: y 21.25 (gym / suite court), x 3.575 (office / gym route and gym), x 4.925 (gym route / suite court). The office's west wall and the south wall's west end stand on the **outward cantilever beams**.
- **Loads and vibration** of the gym (treadmill) and the office over the men's majlis: engineer.

Questions for the owner

1. **Bedroom:** king bed facing the court, or another arrangement (TV wall, sitting area)?
2. **Suite court:** 16.77 m², long and narrow: enough? Clerestory privacy (sill height or obscured glass) to settle.